# Does Money Buy a Longer Life?
# The Preston curve for 158 countries in 2019

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# Introduction

People in rich countries live much longer than people in poor countries. In 2019 a child born in Japan could expect to live 85 years, while a child born in Nigeria could expect 63. In 1975 the demographer Samuel Preston showed that life expectancy rises steeply with income in poor countries and levels off in rich ones [1]. This relation is now called the *Preston curve*. It matters for policy: if extra income buys many years of life in poor countries but only a few in rich ones, then economic growth is most valuable for health where people are poorest.

In this report we ask: **how many years of life expectancy go together with a doubling of a country's income? And is this gain smaller for rich countries than for poor ones?**

# Data sample

We use the Gapminder "Fast Track" dataset [2]. It is a collection of CSV files with one value per country per year, compiled by Gapminder from the UN, the World Bank and other sources. We use three variables:

- `lex`: life expectancy at birth, in years;
- `gdp_pcap`: GDP per capita in international dollars, adjusted for inflation and for purchasing power (PPP), so that incomes can be compared between countries;
- `pop`: population, used to drop very small countries.

We also use the country names and the world region of each country.

We use the year 2019, the last year before the COVID-19 pandemic, which temporarily lowered life expectancy in many countries. We drop countries with fewer than one million inhabitants, because their data are noisier. No further cleaning is needed: all three variables are available for every remaining country. The final sample has **158 countries**. Life expectancy ranges from 53.8 to 85.4 years, and GDP per capita ranges from \$583 to \$120 114.

# Data analysis method

The analysis is a single Python script using `pandas`, `scipy` and `matplotlib`. Plotted against income, life expectancy rises quickly at first and then levels off. Plotted against the *logarithm* of income, the relation is close to a straight line (Figure 1). We therefore fit

$$\text{life expectancy} = a + b \cdot \log_2(\text{GDP per capita}) \tag{1}$$

by least squares, with each country counted once. Using $\log_2$ makes the slope $b$ easy to read: it is the number of years of life expectancy that go together with a *doubling* of income. The uncertainty on $b$ is the standard error from the fit.

To test whether the gain gets smaller for rich countries, we split the sample at the median GDP per capita (\$15 053) and fit each half separately. We then compare the two slopes. The uncertainty on their difference is $\sigma_{\Delta b} = \sqrt{\sigma_{\text{poor}}^2 + \sigma_{\text{rich}}^2}$.

# Results and discussion

The fit gives

$$b = 3.67 \pm 0.18 \text{ years per doubling of income}, \qquad a = 22.8 \pm 2.5 \text{ years}.$$

The correlation is strong: $r = 0.85$, so income alone explains $R^2 = 73\%$ of the differences in life expectancy between countries. The scatter of countries around the line is 3.9 years.

For the poorer half of the countries we find $b_{\text{poor}} = 3.64 \pm 0.45$ years per doubling, and for the richer half $b_{\text{rich}} = 3.46 \pm 0.47$ years per doubling. The difference is $0.18 \pm 0.66$ years, only $0.3\sigma$. The two halves are

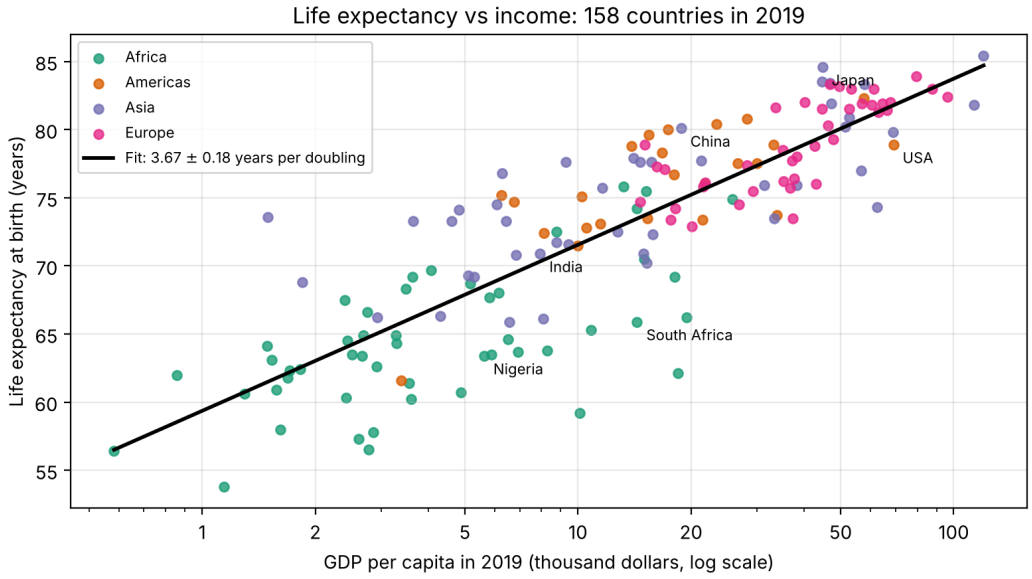



Figure 1: Life expectancy against GDP per capita for 158 countries in 2019. Each dot is a country, coloured by world region. The income axis is logarithmic. The black line is the fit of Eq. (1).

therefore consistent with one and the same slope. In other words, the gain does *not* get smaller per doubling of income.

The curve still flattens in absolute terms, because a doubling costs very different amounts of money. Going from \$1 000 to \$2 000 per person and going from \$50 000 to \$100 000 both add about 3.7 years. The first needs \$1 000 extra, but the second needs \$50 000. So an extra \$1 000 goes together with about 50 times more years of life in a poor country than in a rich one.

Income does not explain everything. China (+5.2 years) and Japan (+5.1) live longer than their income predicts, and the USA lives 2.9 years shorter. The largest negative deviations are in southern Africa, for example Eswatini (−12.4 years) and South Africa (−7.6), where HIV/AIDS strongly lowers life expectancy. They are also found in oil-rich countries such as Equatorial Guinea (−12.7), whose high GDP per capita is not shared widely.

**Conclusion.** Each doubling of income goes together with $3.67 \pm 0.18$ more years of life expectancy. The slope is the same for poor and rich countries within $0.3\sigma$. In dollars, however, the gain is far larger for poor countries, which supports Preston's original finding. Note that this is a correlation, not proof that money causes longer lives: richer countries also have better health care, education and sanitation.

**Use of AI tools.** I used Claude to help write and debug the Python code, and to draft the README and report. I chose the research question, ran the analysis on my own laptop, checked the results, and can explain all steps of the method and the code.

# References

[1] S. H. Preston, *The changing relation between mortality and level of economic development*, Population Studies **29**(2), 231–248 (1975).

[2] Gapminder Foundation, *Gapminder Fast Track dataset*. Available at: `https://github.com/open-numbers/ddf--gapminder--fasttrack` (commit `85af9e1`, accessed October 2026).